BAYU SETIAWAN

SOFTWARE ENGINEER — MACHINE LEARNING & SOFTWARE DEVELOPER

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PROFILE SUMMARY

Fresh Graduate in Computer Science from Universitas Pakuan with a GPA of 3.31, highly dedicated with a strong interest in Software Engineering. Experienced in building data-driven systems for the manufacturing industry, including a Machine Learning-based production prediction system (Random Forest Regression, R^2 0.81 on test data) used to support production planning across 10 product categories. Comfortable working across the full development cycle, from data preprocessing and modeling to integrating models into production-ready web and mobile applications (Flask, Laravel, Flutter). Eager to contribute as a Software Engineer / Data Scientist / Machine Learning Engineer at a product-oriented, data-driven company.

WORK EXPERIENCE

Software Engineer Intern | PT Honoris Industry *August 2025*

Vehicle Fleet Cubication Selection System (Logistics Optimization)

- Designed a mobile-based fleet cubication system using Dart (Flutter) to help operational teams select optimal vehicles for each shipment, reducing unused cargo space and improving fleet selection accuracy.
- Implemented a real-time volume calculation algorithm within the Flutter application to give field operations instant fleet recommendations on-site, cutting manual calculation time and streamlining vehicle selection during shipment.

Machine Learning-Based Production Volume Prediction System

- Built an end-to-end machine learning system to forecast monthly production needs for 10 LED product categories, achieving an R^2 of 0.814 and low MAPE on 86 months of historical test data (Jan 2019 - Feb 2026), using Random Forest Regression.
- Designed a data preprocessing pipeline and tested 4 feature engineering approaches to identify the most business-realistic prediction model, validated with TimeSeriesSplit cross-validation for reliable performance estimation.
- Developed an interactive dashboard to give management real-time visibility into production forecasts and reduce overproduction risk by an average of ~3,900 units/month, built with Flask and integrated automated production-control logic.
- Collaborated with academic advisors and internal company stakeholders to align the model with real operational requirements, ensuring the system was production-ready before deployment.

HI Inventory System Redesign (Web to Mobile Migration)

- Redesigned PT Honoris Industry's HI Inventory system from a web-based platform into a mobile application to improve accessibility and usability for field use, built using Flutter, Laravel, and MySQL.
- Developed a RESTful API to connect the Flutter application with the Laravel backend, enabling real-time data synchronization and centralized inventory data management through MySQL.

ADDITIONAL PROJECTS

- Developed a CRUD-based food ordering website using Laravel to manage menu, orders, and inventory data efficiently.
- Built a personal portfolio website for a client using native PHP to showcase their work and professional profile.
- Redesigned the Bapenda Kota Bogor website using native PHP to improve structure and usability.

EDUCATION

Bachelor of Computer Science — Universitas Pakuan *2022 - 2026*

GPA: 3.31/4.00. Thesis: "Implementation of the Random Forest Algorithm in a Web-Based Lamp Production Information System for Production Quantity Prediction at PT Honoris Industry."

SKILLS

- Programming Languages: Python, JavaScript, PHP, Dart, HTML/CSS
- Frameworks & Tools: Laravel, Flutter, Flask, MySQL, Git/GitHub, Google Colab, VS Code
- Data Science & Machine Learning: Scikit-learn, Pandas, Random Forest Regression, Data Preprocessing, Time Series Cross-Validation, Data Visualization
- Web & Mobile Development: RESTful API, Frontend & Backend Development, Responsive Design
- Soft Skills: Problem Solving, Analytical Thinking, Teamwork, Technical Communication, Time Management